

NGUYEN GIA HUY

Python Backend Engineer

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PROFESSIONAL SUMMARY

Python backend engineer working in core banking: develops and maintains the **Lending** and **Overdraft** products of Vikki Digital Bank as **Python Smart Contracts** on Thought Machine Vault Core, covering the interest, fee, repayment, and settlement lifecycle. Brings nearly three years of backend and data experience, including core banking integrations, AWS ETL pipelines for a utility platform serving millions of users, and test automation.

TECHNICAL SKILLS

Python: Python 3, Vault Core Smart Contracts, FastAPI
Financial Logic: Lending, overdraft, interest accrual, penalty and late fees, repayment, settlement
Backend & APIs: REST/JSON and gRPC APIs, Kafka, event-driven services, Keycloak (RBAC)
Data: SQL, PostgreSQL, Redis, MongoDB, OpenSearch, Amazon Athena
AWS: Lambda, ECS, RDS, S3, EKS
Also used: Golang (testing framework, migration system)

WORK EXPERIENCE

Python Backend Engineer, Core Banking (Middle Software Engineer) 03/2026 – Present
Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

- Developed and maintained the **Lending** and **Overdraft** products as **Python Smart Contracts** on Thought Machine Vault Core, the bank's core ledger.
- Implemented the **interest lifecycle** in contract code, including interest accrual plus penalty and late-fee calculation, driven by Vault Core's contract hooks and **scheduled events**.
- Implemented **repayment and settlement** processing for both products, where every calculation flows directly into customer balances.
- Integrated Vault Core with external backend services that run core banking operations and transaction workflows.
- Rebuilt the product **simulation testing framework** (in Golang, replacing Gauge), cutting test runs from **24 minutes to just over 50 seconds**, and built a 5x faster account migration system.

Software Engineer – Data Processing 04/2025 – 03/2026
Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Built and operated **ETL pipelines** on AWS Lambda, ECS, RDS, S3, and Athena, transforming large-scale utility data for analytics and operational workloads.
- Increased batch processing throughput by **150%** by redesigning pipelines, running work in parallel, and optimizing data access patterns.

Software Engineer 04/2024 – 04/2025
VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Implemented tenant-isolated login and permissions for **20+ customer organizations** with Keycloak authentication and role-based access control.

Blockchain Developer 06/2023 – 09/2023
EsolLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Developed and reviewed **15+ smart contracts** for DeFi and NFT products in Solidity and Move, plus services syncing on-chain events into off-chain databases.

SELECTED PROJECTS

MeetQ | Software Engineer *AI-powered online meeting platform*

- Built a real-time meeting platform on **LiveKit**, integrating **OpenAI models** for live transcription, voice translation, and meeting summarization.
- Designed the **real-time translation pipeline** and optimized conversational context handling to improve translation accuracy and coherence.

DattingQ | Software Engineer *Dating platform*

- Built a gRPC-Gateway, Kafka, MongoDB, and Redis backend: **1.5K requests/second, 99.95% uptime**.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS) B.Sc. Information Technology – Knowledge Engineering

DevOps on AWS Specialization | Coursera